

Week 2 Worksheet (Solutions)

COMP2048

Dr. Kasra Khosoussi

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1 Basic Concepts

Task 1. *Strings and Languages*

Let $\Sigma = \{a, b\}$ and let

$$L = \{w \in \Sigma^* \mid |w| \geq 1, \text{ and } w \text{ starts and ends with the same symbol}\}.$$

- (a) List five strings in L and five strings not in L .
- (b) Describe $\bar{L}$ (the complement of L) in plain English.

Solution.

- (a) **In L :** $a, b, aa, bb, aba, bab, abba, baab$. (Any non-empty string whose first and last characters match.)

Not in L : $\varepsilon, ab, ba, aab, bba, abbb$.

- (b) $\bar{L} = \Sigma^* \setminus L$ consists of two kinds of strings:

- The empty string ε (which is not in L since $|w| \geq 1$ is required), and
- all non-empty strings over $\{a, b\}$ whose first and last characters are *different* (i.e., strings that begin with a and end with b , or begin with b and end with a).

Worth asking students whether $\varepsilon \in L$ before they describe the complement. They should be comfortable with the idea of seeing ε as a string (i.e., $\varepsilon \in \Sigma^*$).

Task 2. *Strings and Languages*

Let $\Sigma = \{a, b\}$, $L_1 = \{w \in \Sigma^* \mid w \text{ starts with } a\}$, and $L_2 = \{w \in \Sigma^* \mid w \text{ ends with } b\}$.

- (a) List five strings in $L_1 \cap L_2$.
- (b) List five strings in $L_1 \cup L_2$.

Solution.

- (a) $L_1 \cap L_2$ contains strings that start with a and end with b .
Examples: $ab, aab, abb, abab, aaabb$.
- (b) $L_1 \cup L_2$ contains strings that start with a or end with b (or both).
Examples: a, ab, bb, aab, bbb .

Make sure students are comfortable with the idea of seeing languages as sets of strings.

2 Deterministic Finite Automata (DFA)

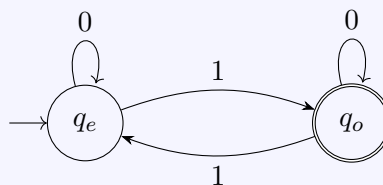
Task 3. *Odd number of 1s*

Let $L = \{w \in \{0,1\}^* \mid \text{the number of 1s in } w \text{ is odd}\}$.

- (a) Construct a DFA over $\Sigma = \{0,1\}$ accepting exactly L .
- (b) Trace on 0110, 111, 1101, and ε .

Solution.

- (a) The DFA has two states tracking the parity of the 1-count: q_e (even, including zero) and q_o (odd). The start state is q_e and the sole accept state is q_o .



(b) **Traces:**

- 0110: $q_e \xrightarrow{0} q_e \xrightarrow{1} q_o \xrightarrow{1} q_e \xrightarrow{0} q_e$. **Reject** (two 1s, even). ✓
- 111: $q_e \xrightarrow{1} q_o \xrightarrow{1} q_e \xrightarrow{1} q_o$. **Accept** (three 1s, odd). ✓
- 1101: $q_e \xrightarrow{1} q_o \xrightarrow{1} q_e \xrightarrow{0} q_e \xrightarrow{1} q_o$. **Accept** (three 1s, odd). ✓
- ε : no transitions taken, stays in q_e . **Reject** (zero 1s, even). ✓

The ε case is a good discussion point (zero is even, so $\varepsilon \notin L$). Make sure they understand the role of q_e and q_o in this example.

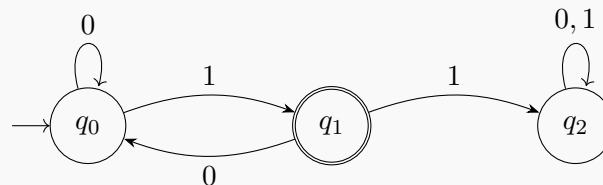
Task 4. *State Diagrams and State Transition Tables*

(i) Transition table for DFA M over $\Sigma = \{a, b\}$, start state q_0 , accept state q_2 .

	a	b
q_0	q_1	q_0
q_1	q_1	q_2
q_2	q_1	q_0

- (a) Draw the state diagram for M .
- (b) Trace M on **ab**, **aab**, and **ba**.
- (c) Describe $L(M)$ (i.e., the language accepted by M) in plain English.

(ii) State diagram for DFA N over $\Sigma = \{0, 1\}$.

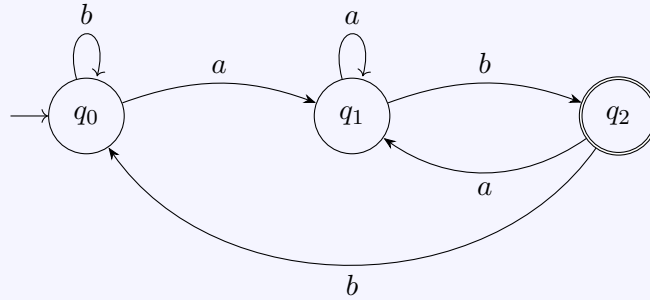


- (a) Write the transition table for N .
- (b) Describe $L(N)$ in plain English.

Solution.

Part (i)

- (a) State diagram for M :



(b) **Traces:**

- **ab:** $q_0 \xrightarrow{a} q_1 \xrightarrow{b} q_2$. Final state $q_2 \in F$ (where F is the set of accepting states; here it only contains q_2). **Accept.**
- **aab:** $q_0 \xrightarrow{a} q_1 \xrightarrow{a} q_1 \xrightarrow{b} q_2$. Final state $q_2 \in F$. **Accept.**
- **ba:** $q_0 \xrightarrow{b} q_0 \xrightarrow{a} q_1$. Final state $q_1 \notin F$. **Reject.**

(c) $L(M)$ is the language of strings over $\{a, b\}$ that end with ab .

Part (ii)

(a) Transition table for N :

	0	1
q_0	q_0	q_1
q_1	q_0	q_2
q_2	q_2	q_2

(b) $L(N)$ is the set of binary strings ending in 1 without having any two consecutive 1s. The machine accepts after reading the first 1 (state q_1); a second 1 sends it to the dead state q_2 from which acceptance is impossible.

Ask students: “What does q_2 represent? Why are there no transitions out of it that lead back to q_1 (i.e., our accepting state)?” This reinforces the idea of a dead/trap state in DFA.